

Answer: A

- 3 Consider the horizontal distance,

$$s_x = u_x t$$

$$10.6 = (12.0 \sin 25.0^\circ) t$$

$$t = 2.1 \text{ s}$$

Consider vertical motion,

$$v_y = u_y + a_y t$$

$$= 12.0 \cos 25.0^\circ + (-9.81)(2.1)$$

$$= -9.7 \text{ m s}^{-1}$$

Speed at which the ball enters the hoop is

$$v = \sqrt{v_y^2 + v_x^2}$$

$$= \sqrt{(-9.7)^2 + (12.0 \sin 25.0^\circ)^2}$$

$$= 10.9 \text{ m s}^{-1}$$

Answer: B